McMaster University | W Booth School of Engineering Practice & Technology

Project: Sushi Assembly Line

Midterm Report

PROCTECH 4TR3: Capstone Design Project II

Automation Systems Engineering Technology

Nicolas Jao (400450151)

Jacob Kwan (400453713)

Jamie Williams (400462359)

Submitted: October 22, 2026

By including our names on the cover, we hereby acknowledge that the work is our own. We also acknowledge the work was done at McMaster. By printing our name on the electronic copy, we acknowledge that this is equivalent to signing our signature on the document.

Abstract

Include here your abstract. It is usually about half a page long.

This is not an introduction to the topic. It contains very concise statements on:

(i) the problem being solved .

(ii) approach you have taken to solve it.

(iii) project outcomes.

(iv) significance of your work.

It does not provide background information on reasons for doing the work. It is often good practice to write this part last, after the report is completed.

Table of Contents

Abstract 2

Table of Contents 3

1 Executive Summary 5

2 Introduction 6

2.1 Motivation 6

2.2 Problem Statement 6

2.3 Project Objectives and Requirements 7

Project Objectives 7

Requirements 7

3 Literature Review 10

4 Design 12

4.1 Original Design 12

4.2 Design Changes 13

4.3 Final Design 15

4.4 Bill of Materials 18

5 Project Implementation 18

5.1 Prototype Description 19

5.2 Group Member Contributions 20

5.3 Functionality Tests 20

6 Discussion 22

6.1 Challenges and How They Were Overcome 22

6.2 Drawbacks and Limitations 23

6.3 Advantages 23

6.4 Proposed Design Changes 24

7 Conclusion and Recommendations..... 24

8 References 25

9 Appendices..... 26

1 Executive Summary

This is the section that your boss' boss will read!

It is a two-page, maximum, summary of your report. It is the section that you want your supervisor to read if they do not have the time to read everything. Through this you should convey the major points of the project, similar to the abstract, but with more detail. State:

- the need/problem you are trying to solve
- the main goals of your project
- the design approach you took
- the results of the implementation
- the benefits of the implementation
- any challenges along the way or anticipated in the future
- any potential future work to be done to the project.

2 Introduction

2.1 Motivation

Sushi is enjoyed around the world, and basic maki rolls are among the most in-demand items for both sushi restaurants and grocery retailers [1], [2]. Although these rolls are simple in design, making them requires several sequential steps (laying out nori and rice, adding fillings, rolling, cutting, and plating), each of which must be done carefully and consistently. The result is a persistent bottleneck in restaurant kitchens and in large-scale grocery production.

This bottleneck shows up most clearly during peak service hours and high-volume production periods. Orders accumulate and preparation times grow. Skilled chefs spend large portions of their shifts on repetitive assembly rather than on the more complex, artistic dishes that make use of their training. In grocery supply chains, slow manual production limits output and makes it difficult to keep up with growing demand for ready-to-eat sushi.

These inefficiencies carry real costs. Restaurants face longer wait times and greater operational strain. Labour costs rise when highly trained staff are tied up in routine tasks. Grocery producers are constrained in how far they can scale. Over time this reduces productivity, increases pressure on staff, and limits growth.

Existing automation only partly addresses the problem. Commercial machines, such as those from AUTEC, automate individual steps like forming rice sheets or cutting rolls, but they still require manual intervention between steps [3]. There is a gap for an integrated system that coordinates the steps of roll production in a single workflow.

2.2 Problem Statement

This project is the design, construction, and programming of a table-scale automated sushi assembly line that produces a cut and plated maki roll with minimal human involvement. After the operator loads the nori, rice, and toppings and presses START, the system flattens the rice onto the nori, places toppings, rolls it, cuts it into even pieces, and plates them. A moving platform on a linear guide rail carries the roll between stations, and a UR3e robotic arm handles the toppings and the final plating. The system is intended to support chefs by taking over the repetitive production of simple rolls, and to serve as a small-scale prototype for larger food manufacturing applications. To

keep the project achievable within the available time, the scope was reduced from the original proposal by simplifying the rice station, using firm toppings such as cucumber or imitation crab, and assigning topping placement to the robotic arm instead of another station with a multi-layered trapdoor contraption.

2.3 Project Objectives and Requirements

Project Objectives

The sushi assembly line shall:

1. Produce uniform pieces of sushi that are consistent in appearance and quality.
2. Run automatically, needing human involvement only to load and replenish materials before each cycle.
3. Be safe for people to work around.
4. Be entirely food-grade so the product is safe to eat.
5. Be modular and easy to clean and service.
6. Display relevant operating information to the user through an HMI.
7. Be quick and efficient compared to making the same roll by hand.

Requirements

System-level

ID	Requirement	Measurable criterion	Verification
S1	Produce one cut roll per cycle	4 pieces per roll	Count pieces after each cycle
S2	Piece uniformity	Each piece 10 mm ± 3 mm thick	Measure all pieces with calipers
S3	Autonomy	0 operator actions between pressing START and a plated roll	Observe 10 cycles
S4	Cycle time	≤ 2 min from START to plated roll	Time 10 cycles

ID	Requirement	Measurable criterion	Verification
S5	Reliability	≥ 80% of rolls stay intact through cutting and plating	10 consecutive trials
S6	Food safety	100% of food-contact surfaces made of food-grade materials	Material review against the BOM
S7	Cleanability	Food-contact parts removable by hand or with basic hand tools in ≤ 10 min	Timed disassembly
S8	Supported toppings	Cucumber or imitation crab strips, about 100 mm long	Test with both toppings

Safety and HMI

ID	Requirement	Measurable criterion	Verification
H1	Emergency stop	Halts all motion (rail, motors, cutter, robot) within 1 s of activation	Trigger during operation at each station
H2	Blade guarding	Blades enclosed or guarded so they cannot be touched during a cut	Inspection
H3	HMI display	Shows the current station and elapsed run time, refreshed at least every 1 s	Observe during a full cycle

Station-level

ID	Station	Requirement	Measurable criterion
0.1	Linear Guide Rail	Length and mounting	1 m long, fastened to the table with ≥ 2 M8 screws
0.2	Linear Guide Rail	Position control	Platform stops at each station within ± 2 mm using limit switches at both ends
1.1	Rice Rolling	Rice thickness	4 mm ± 1 mm across the nori

ID	Station	Requirement	Measurable criterion
1.2	Rice Rolling	Coverage	Rice covers ≥ 90% of the nori area
1.3	Rice Rolling	Roller	Single motor-driven roller, ≤ 30 cm (1 ft) long, driven by one 8–35 V stepper motor, with a non-stick food-safe surface
2.1	Topping Placement	Placement accuracy	Topping line placed within ± 5 mm of the target position, parallel to the roll axis
2.2	Topping Placement	Success rate	Placed correctly in ≥ 8 of 10 attempts
2.3	Topping Placement	Contingency	If 2.2 is not met, a single-layer trapdoor dispenser replaces robotic placement
3.1	Rolling	Platform size	About 15 cm × 13.5 cm
3.2	Rolling	Drive	One independently driven 8–35 V stepper motor
3.3	Rolling	Roll quality	Roll is closed along the full seam and holds its shape through cutting (see S5)
4.1	Cutting	Blades	3 blades at equal spacing, producing 4 pieces
4.2	Cutting	Cut quality	Cut passes completely through the roll with no piece joined to its neighbour
4.3	Cutting	Actuation	Driven by a 8–35 V stepper motor with enough force to complete the cut without crushing the roll
5.1	Plating	Robot and gripper	UR3e arm with a custom gripper that picks up pieces without breaking them
5.2	Plating	Placement	All 4 pieces on the plate in ≥ 8 of 10 cycles

3 Literature Review

Automation of sushi preparation is not a brand-new concept. There exists commercial machines designed to perform specific sushi-making tasks that have been adopted in some restaurants and food service environments. A study from 2025 states that robotic sushi-making systems have achieved a 12% adoption rate across chain restaurants in some form [2]. A notable example in this field is the lineup of commercial sushi machines from AUTEC.



Figure 1: A rice roller machine from AUTEC.
By https://www.sushimachines.com/technical-support/

AUTEC is a company with a long history in sushi automation tools dating back to the 1980s when they released some of the first commercial sushi forming devices and have continued expanding their product range globally [3].

The issue is that they are focused on individual production steps such as creating rice sheets or cutting rolls. This is the current state of sushi machines, or the “prior work” we are building on.



Figure 2: UR3 Universal Robot making coffee.
Screenshot taken from https://www.youtube.com/watch?v=t83wtiVurUI

This project from Mobile Automation demonstrates the capabilities of the UR3 Universal robotic arm as it is used to make a cup of coffee [4]. This provides precedence for how our group intends to use the robotic arm in the last station to pick up the sushi pieces and place them on a plate. Projects like this prove that the UR3 lineup of products can be used for much more complex tasks than our group plans to use it for. However, the use of a robotic arm is simply one part of our intended design.

The gap this project seeks to build on is to move beyond isolated machines that automate *parts* of sushi preparation toward a more integrated assembly line approach. Instead of simply producing rice balls or singular rolls, the proposed system aims to connect and coordinate tasks into a streamlined workflow, reducing the manual intervention still present in existing machines and supporting high-volume production for markets like restaurants and grocery distribution. Our proposed solution will include a full assembly line for a sushi roll from scratch, with multiple substations on a small table with wheels for placing ingredients, rolling the sushi, cutting, and placing them on a plate, ready to serve.

4 Design

4.1 Original Design

The design from PROCTECH 4TR1 was a table-scale, six-station assembly line that would make a maki roll from scratch. The group brainstormed several project ideas in the first week and chose the sushi line because it was a fun concept that would showcase our automation skills in a manufacturing setting. The planned steps were: (1) lay the nori, (2) lay a flat sheet of rice onto the nori, (3) lay the fish and other fillings, (4) roll the sushi, (5) cut the roll into segments and (6) plate the pieces. A UR3e robotic arm, which the group has access to along with its controller, both mounted on a table, was to perform the final pick-and-place plating. The whole line was to be mounted on that same table.

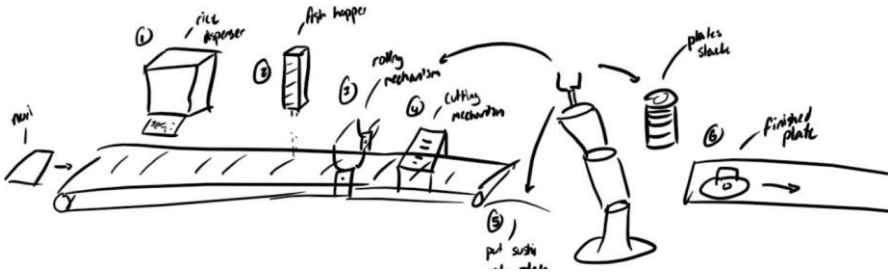


Figure 3: Original sketch of the assembly line and its stations from 4TR1 (rice dispenser, fish hopper, rolling mechanism, cutting mechanism, UR robot plating).

The principle of operation chosen for each station was:

- **Base conveyance:** a linear slide carrying a moving platform. The group first imagined a conveyor belt, then decided on a motor-driven linear slide with the sushi laid on a nori layer on the platform from the very start. This makes the machine a batch process rather than a continuous one, which the group accepted because it avoids the problem of transferring the laid-out ingredients onto the roller. The platform returns to the start for the next batch. The slide was to use a 4040 V-slot extrusion with a belt drive and a stepper motor.
- **Rice dispensing:** inspired by commercial sushi machines, a hopper feeding two counter-rotating rollers that flatten the rice and output it as a sheet.
- **Fish and topping dispensing:** first a wheel with flaps that separated pieces of fish, then (in 4TR1 week 6) a multi-level trapdoor magazine that could preload several batches of fish and other toppings such as avocado and drop them onto the nori as the platform arrived.

- **Rolling:** a "ribcage" roller made of curved ribs on two sides. One side is driven by a motor and the other turns the opposite way through sawtooth gears at the inner end. The ribs close together around a nori mat to roll the sushi.

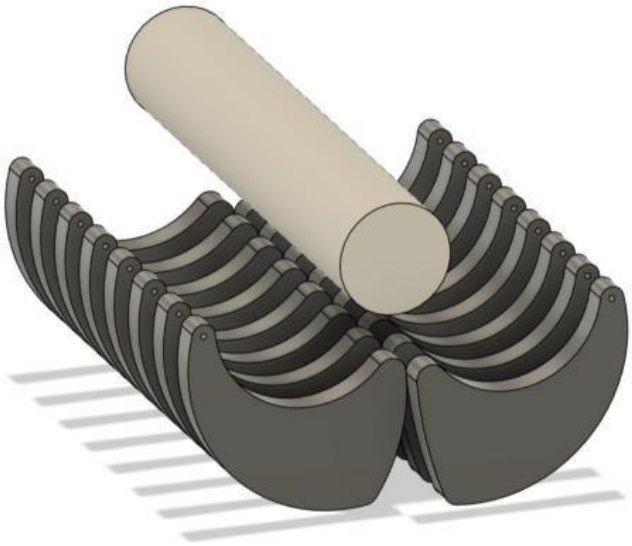


Figure 4: Early CAD prototype of the ribcage roller (4TR1).

- **Cutting:** a guillotine-type blade assembly with several blades mounted in a line that travels straight down through the roll.
- **Plating:** the UR3e picks up the individual pieces and places them on a plate.

The planned control hardware was an Arduino MEGA with a stepper motor driver for the slide, plus DC motors and servo motors at the other stations. A finite state machine defined the sequence IDLE, move forward, station operations, plating and move backwards. Limit switches were to report the platform position at each station.

4.2 Design Changes

The design that is being built in 4TR3 differs from the 4TR1 design in several ways. The changes were forced by two things: testing showed that some original ideas did not work with real food and real parts, and the group reduced the scope after realizing that the project was very ambitious for the time that remained (about 13 weeks when the semester began).

- **Toppings.** Raw fish was replaced with firm toppings such as cucumber or imitation crab, which are much easier to handle as hard solids.

- **Topping station removed.** The multi-level trapdoor dispenser was removed completely. The UR3e now places the toppings. If the robot proves unable to do this reliably, a simple one-layer trapdoor dispenser is the contingency (requirement 2.3).
- **Rice station redesigned.** In week 8 testing with real rice, the rice stuck to the rollers, jammed in the gears and did not squeeze through the gap well. A narrower roller gap and a proposed string scraper only partly helped. The hopper and two-roller station was replaced by a single motor-driven roller across the platform. The platform moves under the roller and the roller flattens the rice onto the nori. The consequence is that the operator now places the rice on the platform before each cycle instead of dumping it into a hopper.
- **Roller material and drive.** The original dollar-store silicone roller pins collect dust and hair, and their shafts spin freely. The group moved to a wooden roller and 3D printed press-fit pins with a flat for the stepper motor shaft. An old 775 DC motor was tried first, but it was far too fast and did not have enough torque, so a 42-40 stepper motor (the same type used elsewhere in the system) was used instead.
- **Motors.** The electrical plan originally listed DC and servo motors at several stations. In week 8 the group decided to use the stepper motors it already had by default, to save money, and to buy different motors only where torque proved insufficient.
- **Ribcage roller.** The first prototype used two motors so that the order in which the ribs close could be controlled, and the motors were moved onto the platform so that no wires fly as the platform moves. Over the summer the printed roller warped and its gears no longer lined up. The roller was redesigned (V2) with the two sides tied together by gears and driven by a single motor, with a tighter closing gap, a smaller inner radius, integrated pillars, more infill and fewer ribs. The reduced rib count leaves room for three blades that cut the roll into four pieces. The redesign also aims to close the roll tighter so that the nori overlaps and seals.
- **Position feedback.** Limit switches only work from one direction, and a set of bidirectional-looking roller switches ordered in March did not work because the motor stalled before pushing them. Limit switches remain at the two ends of the rail. Reed switches, triggered by a magnet on the platform, now mark the two stops in the middle of the line (end of rice rolling and robot topping placement).
- **Cutter.** The guillotine idea was developed into a frame of parallel single-edge blades that is driven vertically by a lead screw and stepper motor, reusing the Z-axis design of the Ender 3 V2 3D printer that many of our parts came from. Clear acrylic side guards were added for safety.

Belt drive, cutting wires, spinning saw blades and a knife held by the robot were considered and rejected.

- **Robot gripper.** A commercial gripper was investigated through Universal Robots, which referred the group to a third-party OEM (Advanced Motion & Controls Ltd.). The expected cost was out of reach of the roughly $400 student budget, so the group will design its own gripper using 3D printed parts and servo motors.

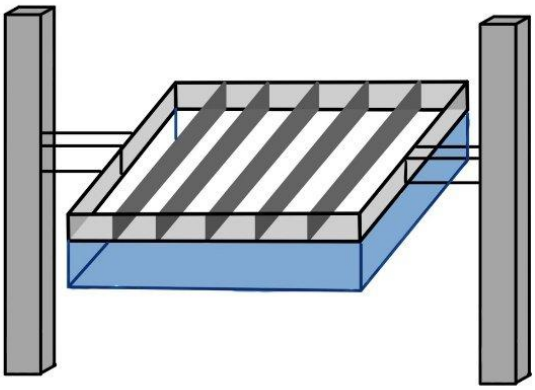


Figure 5: Concept sketch of the cutting station: parallel blades in a square frame lifted by a lead screw mechanism (not shown in image), with acrylic guards in blue.

The 4TR1 table of assembly line steps and operating principles was updated accordingly. The sequence of the final line is: the operator loads the nori and rice on the platform, the platform moves under the rice roller, the platform stops for the robot to place toppings, the ribcage roller rolls the sushi, the cutter slices it into four pieces, and the UR3e plates them.

4.3 Final Design

The final design is a set of modules mounted on an aluminum table with a grid of M8 threaded holes spaced 55 mm apart. A single master assembly in Onshape contains the table, the UR3e robot, and all of the stations, so that the layout and clearances could be planned before anything was built.

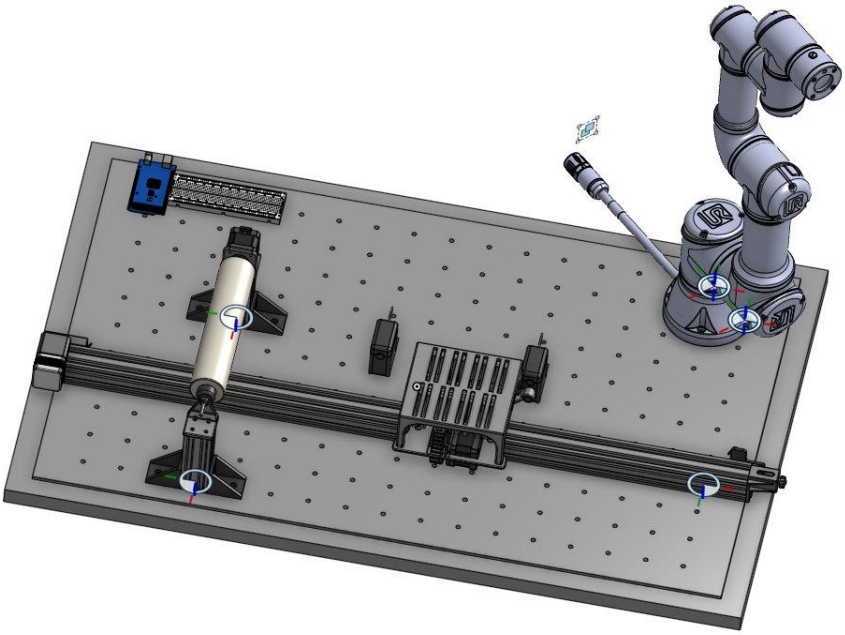


Figure 6: Master assembly CAD of the table, linear slide, rice roller station, reed switch mounts and UR3e robot (October 2026).

Station 0: Linear slide base. A 1 m 4040 V-slot rail carries a 3D printed platform on wheels. The platform is pulled by a GT2 belt around a pulley driven by a 42-40 (JK42HS34-0844 NEMA 17) stepper motor, run through an A4988 motor driver from an Arduino MEGA. A limit switch is located at each end of the rail. The rail is bolted to the table with M8 bolts through two holes drilled through the rail, with access holes on top for tightening the bolts. The platform baseplate has a holder for the actuator magnet of the reed switches and mounting holes for the ribcage roller platform.

Station 1: Rice rolling. A wooden roller sits on 2020 T-slot extrusion uprights above the platform, with a gap of 8 mm between the roller and the platform to compress the rice. A 42-40 stepper motor turns the roller through a 3D printed pin that is a press fit on both the roller and the motor shaft. The motor is held in a 3D printed mount, and the uprights are fastened to the table with 3D printed L brackets and M8 bolts.

Station 2: Robot topping placement. The UR3e is bolted to the table and will move to place cucumber or imitation crab strips on the rice while the platform is stopped at the second reed switch. The gripper will be a custom 3D printed design using an MG996R servo mounted to the robot flange (section 7.11 of the UR manual covers end effector integration). The gripper design includes two claws on gears driven by one servo motor and a servo horn, all encased by 3D printed parts that directly attach to the flange.

Station 3: Ribcage roller. The V2 ribcage roller is mounted on an elevated platform on top of the sliding platform. Two sets of ribs close around a nori mat and are synchronized through gears and driven by one stepper motor. The 3D printed components are the top, ribs, pins, gears, and base.

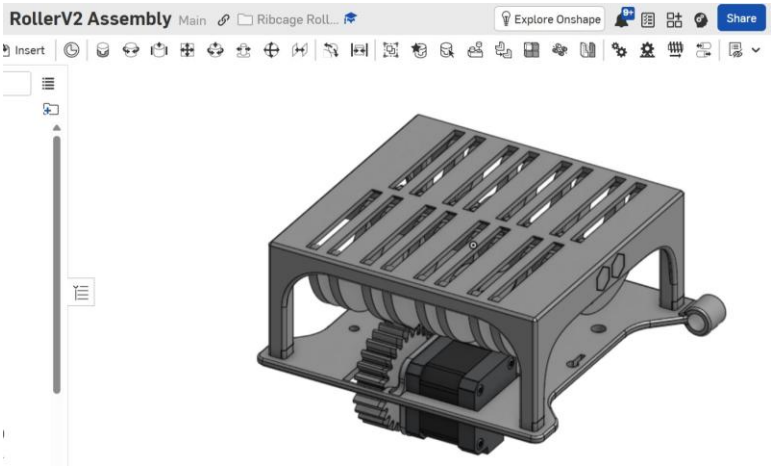


Figure 7: Ribcage roller V2 assembly (CAD).

Station 4: Cutter. The cutting station is at the design stage. A frame carrying three parallel straight blades, spaced to produce four pieces, will be moved vertically by a lead screw and stepper motor taken from the Creality Ender V3 3D printer Z-axis design, between two 2020 extrusions spaced about 250 mm apart. Clear acrylic panels will protect the user from the blades.

Position sensing. Reed switches in 3D printed mounts, bolted to the table beside the rail, report when the platform reaches each intermediate station. The mount is a two-part housing (a base and a snap-on lid) with an L bracket lip that is bolted down through two holes, and the reed switch is slid into the slot.

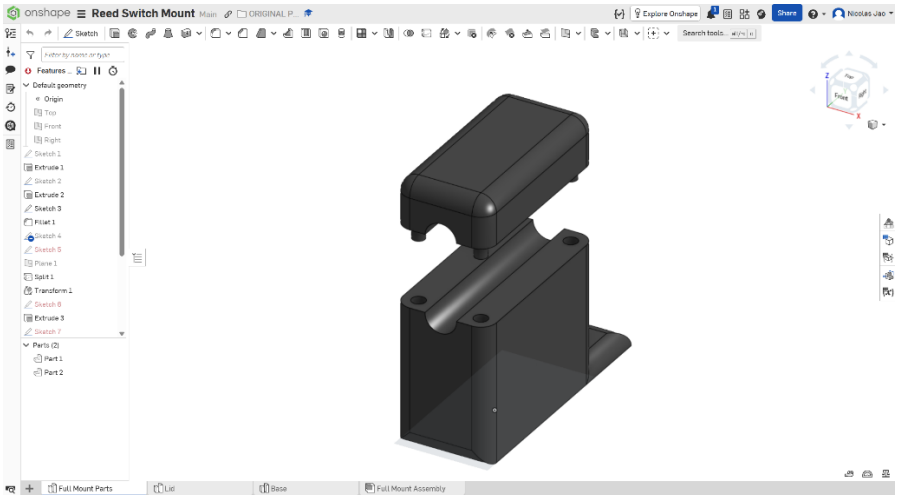


Figure 8: Reed switch mount design in Onshape.

Control system. An Arduino MEGA runs the sequence as a finite state machine, with a stepper driver for each motor. The final wiring layout and the number of controllers is still to be decided. The UR3e and the Arduino must exchange "your turn" signals. The planned approach is to use the digital I/O of the robot controller with an externally powered 24 V relay or interface circuit between the two, and the gripper servo will be controlled by the same Arduino controller.

4.4 Bill of Materials

Include here a list of the materials you used. If the materials were purchased for this project, include prices (in CAD) and links to purchase sites.

Note about Tables:

- Captions on Tables go to the top of them. All Tables should be numbered sequentially.
- Please, do not leave huge hyper links on the Table (on anywhere else). Right click on them and edit what shows up on the page (the link will still work).

Table 1: Bill of Materials. Values are in CAD.

Description	Quantity	Unit Price	Total Price
Arduino Mega https://www.digikey.ca/en/products/detail/arduino/A000067/2639006	1	73.55	73.55
Nema 17 Stepper Motors Amazon Link	4	18.99	75.96
Breadboards Amazon Link	1	9.99	9.99
		TOTAL	159.5
		TOTAL with HST and Shipping	**180.23**

*BOM WILL BE COMPLETE FOR FINAL REPORT

5 Project Implementation

The prototype that exists so far consists of the completed linear slide base (Station 0) and the assembled rice rolling station (Station 1), together with the ribcage roller prototype and reed switch position

sensing that have been tested. The cutting station and the robot gripper are designed or being designed but have not been built.

5.1 Prototype Description

The linear slide is shown below. The 1 m rail carries the platform, which is moved by the belt and stepper motor and stopped at either end by limit switches. The rice rolling station stands over the platform and is driven by its own stepper motor. The reed switch mounts are bolted to the table beside the rail and detect the magnet on the platform.



Figure 9: The assembled linear slide base with its stepper motor and limit switch.

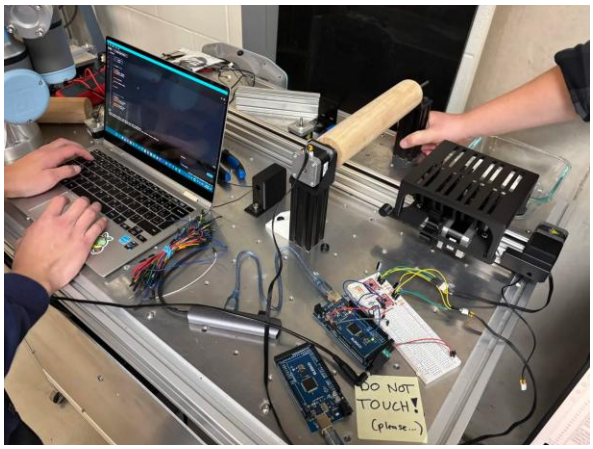


Figure 10: Assembly and testing of the rice rolling station and ribcage roller on the linear slide.

5.2 Group Member Contributions

Nicolas Jao set up the project tools (Google Drive, GitHub repository, Onshape, OneNote and MS Teams), chose the controller and counted the motors and drivers for the electrical bill of materials, and maintained the BOMs. He created and maintained the master assembly CAD, including the table hole grid, which he derived from the UR3e base mounting drawing, and the imported parts. He designed the reed switch mounts and the L brackets for the extrusions, researched fasteners and connectors and sourced and ordered the rail, belt and other parts, and drilled the rail with the group. He works on the UR3e gripper and robot integration, and wrote logbook entries and parts of this report.

Jacob Kwan programmed and tested the stepper motors and drivers with the Arduino, wrote the limit switch and reed switch control programs, and led the troubleshooting of the controller and motors. He made the state diagram of the system, researched the limit switches, created the project plan and schedule, communicated with Universal Robots for the gripper quote and helped work on the midterm presentation.

Jamie Williams designed, printed and redesigned the ribcage roller (both the first prototype and V2), supplied the 3D printer parts used for the linear slide, designed the motor mounts and the gears that connect the motors to the rollers, and printed the platform baseplate, reed switch mounts and L brackets. He is designing the frame for the cutting station.

All three members took part in assembly, the design discussions and the testing sessions.

5.3 Functionality Tests

Linear slide. The stepper motor first only vibrated and made noise on the bench test with the Arduino MEGA, because the supply voltage was too low. After raising the supply voltage and cleaning up the code, it moved smoothly and the limit switch reversed the platform. After assembly of the full slide in March, the motor drove the platform at a speed suitable for the line and the platform reversed when it touched the end limit switches. Bolting the rail down to the table with M8 bolts held tightly.

Reed switches. The first two-way roller limit switches were tested on the slide and failed because the motor stalled before the switch was pushed. The reed switches were then tested with the platform, and the control program stopped the platform correctly at each one. The first printed mounts had dimensional errors, so the mount was redesigned (holes moved and sized for a tighter fit) and reprinted.

Rice rollers (original design). Rice was tried on the original two-roller design in week 8. The rice was very sticky and was not squeezed through evenly, and it stuck to the rollers and jammed in the gears. After the roller gap was narrowed the sheet of rice was cleaner but still stuck to the rollers. This testing led to the single-roller redesign.

Ribcage roller. The first prototype was tested with nori, rice and playdough as a stand-in for toppings, with the motors driven by Jacob's Arduino program. The roller rolled the sushi into the specified diameter, but the nori did not overlap at the top to seal the roll, and the roll opened slightly when the ribs lifted. A play-dough roll made earlier with the nori mat and plastic wrap also showed the roll could be formed. The V2 roller was designed to fix this, and it was assembled and tested for fit with the rest of the station. Jamie found one more change was needed because the gears had no room to turn freely, so a new version must be printed.

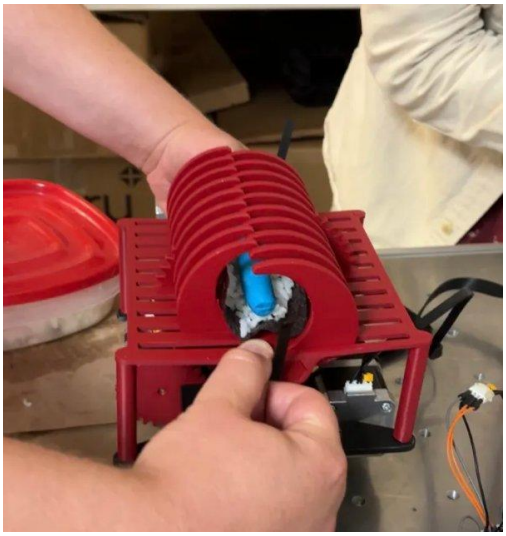


Figure 11: Testing the first ribcage roller prototype with nori, rice and a play-dough topping.

Rice rolling station. The roller station was assembled and tested in week 14 and again in week 15. The 8 mm roller-to-platform gap was a best estimate, but it appears to be right. The roller flattened the rice as intended. Rice and nori were not yet placed in the week-15 test, so the station was only tested with the motor and platform movement. A 775 DC motor tried earlier was too fast and too weak to turn the roller, so the stepper motor replaced it and showed enough torque.



Figure 12: Rice being flattened onto nori by the roller station.

Not yet tested. The cutting station, the UR3e topping placement and the gripper, the robot and Arduino interface, and the full end-to-end cycle have not been built or tested.

6 Discussion

6.1 Challenges and How They Were Overcome

The biggest challenges were practical. The flood in the engineering building in February closed the lab for a time, and the group worked from the library. Getting the dimensions and thread size of the table holes took weeks, until the professor provided the robot base drawing and the group worked backwards from its 55 mm hole spacing. The threads were later confirmed as M8.

Mounting parts was a recurring problem. The M8 bolts did not fit the rail groove and the smaller T-nuts passed through the table holes, so two holes were drilled through the rail, with access holes added on top after it turned out an Allen key could not reach from the side. Spare T-slot extrusions found in the lab did not match the V-slot rail, so they were reused for the other stations. Printed parts also caused problems: the Onshape Hole Wizard threads do not exist as real geometry in a print, so the sensor mount was changed to an L bracket lip, and the first reed switch mounts had dimensional errors.

Sourcing and parts failures cost time. The rail did not arrive until after the session when it was needed, the rail had no threaded holes for the motor mount, and the bidirectional limit switches, the 775 motor and the motor driver from our Arduino kit (which needed a higher supply voltage) all failed to work.

Motor drivers were also damaged by short circuits. The group replaced them with cheap parts and by changing the design (reed switches, a stepper motor). Little project work was done over the summer and the printed ribcage roller warped in the meantime, so the group began the semester by reducing the scope and writing a schedule.

Food-related behaviour was hard to predict. Sticky rice jammed the rollers, and the first ribcage roller did not seal the nori. These are being addressed by the redesigns described above. Food-grade materials were researched (nylon, stainless steel and acrylic) but not yet applied.

6.2 Drawbacks and Limitations

- The system is a batch process, not continuous, and the operator must place the nori and the rice on the platform before every cycle.
- Several 3D printed components (PLA) are not food-grade and are difficult to clean. Requirements S6 and S7 are not yet met.
- The cutter and robot topping placement are still to be built, so the full cycle has not been demonstrated.
- The Arduino-to-robot interface and the gripper are not designed in detail. The UR3e has a limited reach, so it may have to be moved on the table to reach the platform stop where toppings are placed.
- The budget is small (about $400 from the faculty), so the design leans on parts from an old 3D printer and spare lab stock, which limits performance and part quality.

6.3 Advantages

- The design is modular. Each station is on its own mount on the table's 55 mm hole grid, so stations can be moved, replaced or serviced independently.
- Building on spare 3D printer parts and lab stock (extrusions, steppers, bolts, belts) kept costs low, and using one type of stepper motor at most stations simplifies spares, drivers and code.
- The linear slide with a belt drive gives simple, repeatable motion, and reed switches give the position feedback without any force on the platform.
- Having all stations in one Onshape master assembly lets the group check fit and clearances before printing. 3D printed parts are fast to modify when testing shows a problem.

- Because the line is a complete workflow (from rice to plated pieces) and not a single machine, it goes beyond the single-step commercial machines described in the literature review.

6.4 Proposed Design Changes

- **Functionality:** finish the V2 ribcage roller so the nori overlaps and seals; add a nori laying and rice dispensing step so the operator is not needed after START; and consider a stronger motor or gearing for the rice roller.
- **Manufacturability:** replace critical PLA parts with acrylic (laser cut), nylon or stainless steel where they hold their shape better and are food-safe; standardize the fasteners and standardize the extrusion types; and replace hand-drilled holes with machined ones (the machine shop in the building can cut extrusions).
- **User access and safety:** add the acrylic blade guards, an emergency stop and an HMI, and make the food-contact parts easy to take off by hand for cleaning.
- **Control:** consolidate the wiring on a single labeled board, and replace the temporary Arduino and A4988 driver setup with a more robust driver if needed.

7 Conclusion and Recommendations

Conclusions should relate the project goals, stated in the Introduction, with the outcomes of your work.

- Have the goals been achieved? How was that measured?
- Does the functionality of the prototype meet the expectation with respect to the usability?

Recommendations should include you view of possible improvements to your work as well as recommendations for possible applications of your solution in real life

8 References

AUTEC. (2026). *Our Story*. Retrieved March 2, 2026, from https://www.sushimachines.com/cd/about/

Automation, M. (2017, June 21). *YouTube*. (Mobile Automation) Retrieved April 7, 2026, from https://www.youtube.com/watch?v=t83wtiVurUI

FoodSafety. (n.d.). *Safe Selection and Handling of Fish and Shellfish*. (FoodSafety.gov) Retrieved 3 2, 2026, from https://www.foodsafety.gov/blog/safe-selection-and-handling-fish-and-shellfish

Market Reports World. (2025). *Sushi Restaurants Market Size, Share, Growth, and Industry Analysis, By Type (Traditional Sushi Restaurant,Conveyor Belt Sushi Restaurant), By Application (Dine-in,Takeout), Regional Insights and Forecast to 2034.* Market Reports World.

Microsoft. (2025). *Add Citations in a Word document*. Retrieved 2025, from Microsoft Support: https://support.microsoft.com/en-us/office/add-citations-in-a-word-document-ab9322bb-a8d3-47f4-80c8-63c06779f127

Technavio. (2024). *Sushi Restaurants Market Growth Analysis - Size and Forecast 2025-2029.* Technavio.

Varsamopoulos, G. (2004). *How to Write a Technical Paper: Structure and Style of the Epitome of your Research.* Retrieved from https://www.researchgate.net/publication/321686715_How_to_Write_a_Technical_Paper_Structure_and_Style_of_the_Epitome_of_your_Research

9 Appendices

Material which is too detailed to be included in the body of the report should be presented here. All the information included here must be mentioned previously in your report and referred to this section. Place here any information that takes away from the flow of the report (e.g. complicated schematics that were referenced in your report, reference documentation, ancillary materials).